

International Islamic University Chittagong

Department of Computer Science and Engineering



A Thesis Proposal on:

Lightweight Transfer Learning-Based Plant Disease Classification from Real-World Leaf Images

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Contents

1. Introduction
2. Motivation and Problem Statement
3. Applications
4. Research Objectives
5. Challenges
6. Literature Review
7. Research Gaps
8. Methodology
9. Dataset with Sample Data per Class
10. Expected Results
11. System Requirements
12. Work Timeline
13. Conclusion
14. References

1. Introduction

1. Plant diseases can reduce crop quality and yield if they are not identified early.
2. Manual disease diagnosis depends on expert observation, which may be slow, costly, and difficult for large farms.
3. Computer vision and deep learning can classify plant diseases from leaf images, but many existing studies perform well mainly on clean laboratory-style datasets.
4. This proposal focuses on two public real-world datasets: PlantDoc and PlantWild.
5. The goal is to build and compare lightweight transfer-learning models that can work better on noisy real-world plant disease images.

2. Motivation and Problem Statement

- Many plant disease models show very high accuracy on clean datasets such as PlantVillage, but their performance decreases on real-world images.
- Real-world images contain complex backgrounds, different lighting conditions, blur, shadows, and disease spots of different sizes.
- PlantDoc is small and challenging, while PlantWild is larger and collected for in-the-wild plant disease recognition.
- A practical plant disease model should not only be accurate; it should also be small, fast, and suitable for mobile or low-resource devices.

Problem statement

- Existing models often fail to generalize well from clean plant images to real-world plant disease images.
- There is a need for a lightweight and explainable model that can classify plant diseases on PlantDoc and PlantWild with improved robustness.

3. Applications

- Mobile-based plant disease diagnosis for farmers and agricultural workers.
- Early disease detection to reduce crop loss and support timely treatment.
- Low-cost AI support for rural or resource-limited agricultural areas.
- Potential integration with IoT, drones, or smart farming systems.
- Explainable output using Grad-CAM can help users see which leaf region influenced the prediction.

4. Research Objectives

4.1 Primary Objective

To develop a lightweight transfer-learning-based plant disease classification system using PlantDoc and PlantWild datasets.

4.2 Secondary Objectives

- ❖ To prepare and analyze PlantDoc and PlantWild datasets for classification experiments.
- ❖ To train baseline models such as MobileNetV3, EfficientNet-B0/B1, and ResNet50.
- ❖ To improve performance using data augmentation, class balancing, and preprocessing.
- ❖ To evaluate models using accuracy, precision, recall, F1-score, macro-F1, confusion matrix, model size, and prediction time.
- ❖ To use Grad-CAM to explain whether the model focuses on disease regions instead of irrelevant background.
- ❖ To compare performance on PlantDoc and PlantWild and analyze generalization on real-world plant disease images.

5. Challenges

Challenge	Why it matters	Possible solution
Small dataset size in PlantDoc	Deep models may overfit and memorize images.	Use transfer learning, augmentation, and early stopping.
Class imbalance	Some classes may have fewer images, causing poor minority-class performance.	Use class weights, focal loss, oversampling, and macro-F1.
Complex background	Model may learn background instead of disease symptoms.	Use cropping, background reduction, and Grad-CAM checking.
Lighting and camera variation	Same disease can look different in different images.	Use brightness/contrast augmentation and normalization.
Similar disease symptoms	Leaf spot, blight, rust, and mildew can look visually close.	Use stronger feature extractors and analyze confusion matrix.
Limited computation	Large models may not be practical for field devices.	Use lightweight models and compare model size/inference time.

6. Literature Review

The following studies are selected because they are related to real-world plant disease classification, PlantDoc, PlantWild, lightweight models, and generalization problems.

No	Title / Author / Year	Dataset and Model	Result / Accuracy	Limitation / Relevance
1	PlantDoc: A Dataset for Visual Plant Disease Detection Singh et al., 2019/2020	Dataset: PlantDoc Task: plant disease classification/detection baseline	Introduced PlantDoc with 2,598 images, 13 plant species, and up to 17 disease classes.	Small and noisy real-world dataset; good source dataset for this thesis.
2	Benchmarking In-the-Wild Multimodal Plant Disease Recognition and A Versatile Baseline Wei et al., 2024	Dataset: PlantWild, also benchmarked PlantDoc and PlantVillage Model: CLIP/ViT-based MVPDR	PlantWild introduced as a large in-the-wild plant disease dataset with image and text information.	Useful for real-world generalization; multimodal approach is more advanced than basic CNN.
3	Evaluation of Lightweight and Efficient Deep Learning Models for Plant Disease Classification Duhan et al., 2026	Datasets: PlantDoc, Mango, Soybean Models: MobileNetV2, MobileNetV3Small, ShuffleNetV2, YOLOv8n, YOLOv11n	MobileNetV2 reported 89.86% accuracy on augmented PlantDoc.	Shows lightweight models and augmentation can improve PlantDoc performance.
4	Plant Leaf Disease Detection Using Deep Learning: A Multi-Dataset Approach Krishna et al., 2025	Datasets: PlantDoc + web-sourced images Models: EfficientNet-B0, EfficientNet-B3, ResNet50, DenseNet201	EfficientNet-B3 reported 73.31% on PlantDoc and 80.19% on combined dataset.	Shows real-world dataset generalization is still difficult.
5	Plant Disease Classification in the Wild Using Vision Transformers and Mixture of Experts Salman et al., 2025	Datasets: PlantVillage and PlantDoc Model: Vision Transformer + Mixture of Experts	Reported around 74% accuracy on PlantDoc and 68% cross-domain PlantVillage to PlantDoc.	Transformer model is strong but computationally heavier.
6	A Robust Ensemble Model for Plant Disease Detection Using Deep Learning Architectures Zubair et al., 2025	Datasets: PlantVillage, PlantDoc, FieldPlant Models: Ensemble of InceptionResNetV2, MobileNetV2, EfficientNetB3	Reported 99.69% on PlantVillage, 60% on PlantDoc, and 83% on FieldPlant.	Useful evidence that PlantDoc is harder than clean datasets.
7	SerpensGate-YOLOv8: An Enhanced YOLOv8 Model for Accurate Plant Disease Detection Miao, Meng, and Zhou, 2025	Dataset: PlantDoc Model: Enhanced YOLOv8 for object detection	Reported mAP@0.5 = 64.9% and precision = 0.719.	Detection study, not pure classification; still useful for PlantDoc difficulty.
8	PND-Net: Plant Nutrition Deficiency and Disease Classification Using Graph Convolutional Network Bera et al., 2024	Datasets: Banana, Coffee, Potato, PlantDoc Model: CNN backbone + Graph Convolutional Network	Reported 84.30% accuracy on PlantDoc using Xception backbone.	More complex than simple lightweight transfer learning.

7. Research Gaps

- Many studies still depend heavily on clean datasets, while real-world datasets remain more difficult.
- PlantDoc has limited image size, class imbalance, background noise, and lighting variation.
- PlantWild is larger and more realistic, but it also introduces higher class diversity and intra-class variation.
- Many high-performing methods use heavy models or complex transformer/ensemble architectures.
- There is a gap for a simple lightweight method that compares PlantDoc and PlantWild while also using explainability.

Proposed research gap

- This thesis will focus on lightweight transfer learning with augmentation, class balancing, preprocessing, and Grad-CAM explainability for real-world plant disease classification using PlantDoc and PlantWild.

8. Methodology

PlantDoc + PlantWild -> Preprocessing -> Split -> Baseline Models -> Improvement Techniques -> Evaluation -> Grad-CAM Explainability

1. Data collection: Download PlantDoc and PlantWild from public sources and organize folders according to class labels.
2. Data inspection: Check total images, class names, class distribution, duplicates, corrupted files, and image quality.
3. Preprocessing: Resize images, normalize pixel values, and optionally crop leaf regions or reduce background noise.
4. Dataset split: Create train, validation, and test sets while keeping class distribution balanced where possible.
5. Baseline training: Train pretrained lightweight CNN models such as MobileNetV3 and EfficientNet-B0/B1; include ResNet50 as a stronger baseline.
6. Improvement techniques: Apply augmentation, class weights/focal loss, oversampling, and preprocessing improvements.
7. Cross-dataset analysis: Train and test within each dataset and optionally test whether the model trained on one dataset generalizes to the other.
8. Evaluation: Compare accuracy, precision, recall, F1-score, macro-F1, confusion matrix, model size, and inference time.
9. Explainability: Use Grad-CAM to visualize the image regions used by the model for prediction.

9. Dataset with Sample Data per Class

9.1 Dataset Overview

Dataset	Type	Size / Classes	Access	Reason for use
PlantDoc	Real-world visual plant disease dataset	2,598 images; 13 plant species; up to 17 disease classes / about 27 total categories including healthy classes	Official GitHub / Kaggle versions	Small, noisy, challenging, and suitable for improvement-based thesis.
PlantWild	In-the-wild multimodal plant disease dataset	18,542 images; 89 classes according to the Hugging Face dataset page	Official project page / Hugging Face / Google Drive	Larger real-world dataset for stronger evaluation and generalization analysis.

9.2 Sample Data per Class

- ❖ After dataset download, random sample images from every class will be visualized to verify label quality and class variation.
- ❖ The table below shows example class names that can be used during sample visualization. The final class list will be confirmed after downloading the exact dataset versions.

Dataset	Example sample classes / categories
PlantDoc example classes	Apple Black Rot, Bell Pepper Bacterial Spot, Blueberry Healthy, Cherry Powdery Mildew, Corn Gray Leaf Spot, Grape Black Rot, Potato Early Blight, Tomato Leaf Mold, Tomato Late Blight
PlantWild example categories	Multiple in-the-wild crop-disease categories across 89 classes; final class names will be extracted from the downloaded PlantWild label file/folder structure.

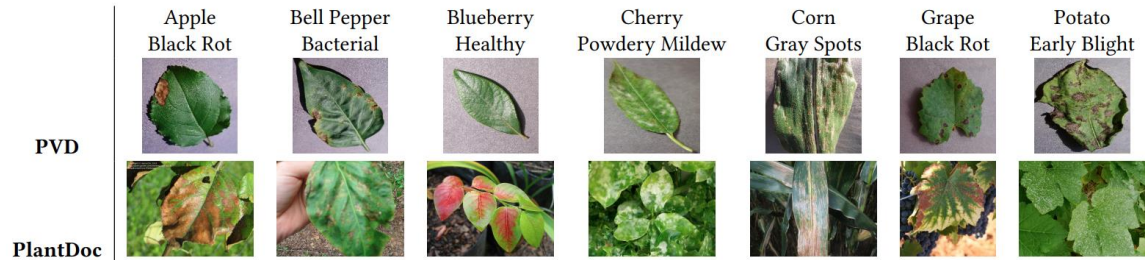


Figure 1: Example PlantDoc samples showing differences between controlled images and real-world images.

10. Expected Results

- ✓ A trained lightweight model for real-world plant disease classification using PlantDoc and PlantWild.
- ✓ A clear comparison of baseline and improved models.
- ✓ Evidence of how augmentation, class balancing, and preprocessing affect performance.
- ✓ Evaluation using accuracy, precision, recall, F1-score, macro-F1, confusion matrix, model size, and inference time.
- ✓ Grad-CAM visualizations to explain whether the model focuses on diseased leaf regions.
- ✓ A recommendation of the best lightweight model for practical plant disease classification.

11. System Requirements

Requirement	Details
Programming language	Python
Development platform	Google Colab / Jupyter Notebook
Deep learning framework	TensorFlow/Keras or PyTorch
Useful libraries	OpenCV, scikit-learn, NumPy, pandas, matplotlib
Hardware	Google Colab GPU such as T4, or local GPU if available
Evaluation tools	Classification report, confusion matrix, Grad-CAM, model size and inference-time measurement

12. Gantt Chart

Task	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Literature review and proposal refinement	■	■				
Dataset download and organization	■	■				
Data analysis and preprocessing		■	■			
Baseline model training		■	■			
Augmentation and class balancing			■	■		
Preprocessing improvement and cross-dataset testing			■	■	■	
Evaluation and Grad-CAM explainability				■	■	
Result analysis and report writing					■	■
Final thesis revision and presentation						■

Mapping the Research to Complex Engineering Problem Attributes

Title of the Project/Thesis: Lightweight Transfer Learning-Based Plant Disease Classification from Real-World Leaf Images

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Which attributes of the Complex Engineering Problems (P) are present?

Attribute of Complex Engineering Problem	Short Explanation
P1: Depth of knowledge required	The study needs knowledge of image classification, CNNs, transfer learning, preprocessing, augmentation and evaluation metrics.
P2: Range of conflicting requirements	The model should be accurate, but it also needs to be lightweight, fast and practical for field or mobile use.
P3: Depth of analysis required (no obvious solution)	There is no single obvious solution because PlantDoc and PlantWild images are noisy, imbalanced and collected in real-world conditions.
P4: Familiarity of issues (infrequently encountered issues)	Unlike clean lab datasets, these datasets contain messy backgrounds, lighting changes, blur and similar disease symptoms.
P5: Extent of applicable codes	The work follows standard ML practice: fair data split, reproducible training, public dataset use and proper evaluation.
P6: Stakeholder involvement and conflicting requirements	Farmers, researchers and field users may need different things, such as high accuracy, fast prediction and simple use.
P7: Interdependence	Dataset quality, preprocessing, augmentation, model choice and evaluation are connected; changing one step affects the final result.

Which attributes of Complex Engineering Activities (A) are involved?

Attribute of Complex Engineering Activity	Short Explanation
A1: Range of resources	The project uses public datasets, Python, Google Colab, deep learning libraries, pretrained models and research papers.
A2: Level of interactions	The work needs regular supervisor feedback and clear communication of methods, results and limitations.
A3: Innovation	The study combines lightweight models with augmentation, class balancing and Grad-CAM for real-world disease classification.
A4: Consequences to society and the environment	Early disease detection can help reduce crop loss and may reduce unnecessary pesticide use.
A5: Familiarity	This is not a routine image task because the images are noisy, inconsistent and harder than clean datasets.

Comments:

This thesis addresses a complex engineering problem because it works with real-world plant disease images, limited and imbalanced data, and a trade-off between accuracy and model efficiency. The study compares models, improves training reliability and explains predictions for practical agricultural use.

Approval

Approval of Supervisor	Approval of Examination Committee
Signature	Signature

13. Conclusion

13.1 Key Takeaways

- Plant disease classification on real-world images is more challenging than classification on clean laboratory images.
- PlantDoc provides a small and noisy dataset, while PlantWild provides a larger in-the-wild dataset for stronger evaluation.
- Lightweight transfer learning can make the proposed system more practical for mobile or low-resource environments.

13.2 Impact and Future Work

- The proposed work may support early disease diagnosis and reduce crop loss.
- Future work can include mobile deployment, disease severity estimation, object detection, segmentation, or multilingual farmer-support applications.
- Further improvement may be achieved by adding more field images and expert-verified labels.

14. References

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